

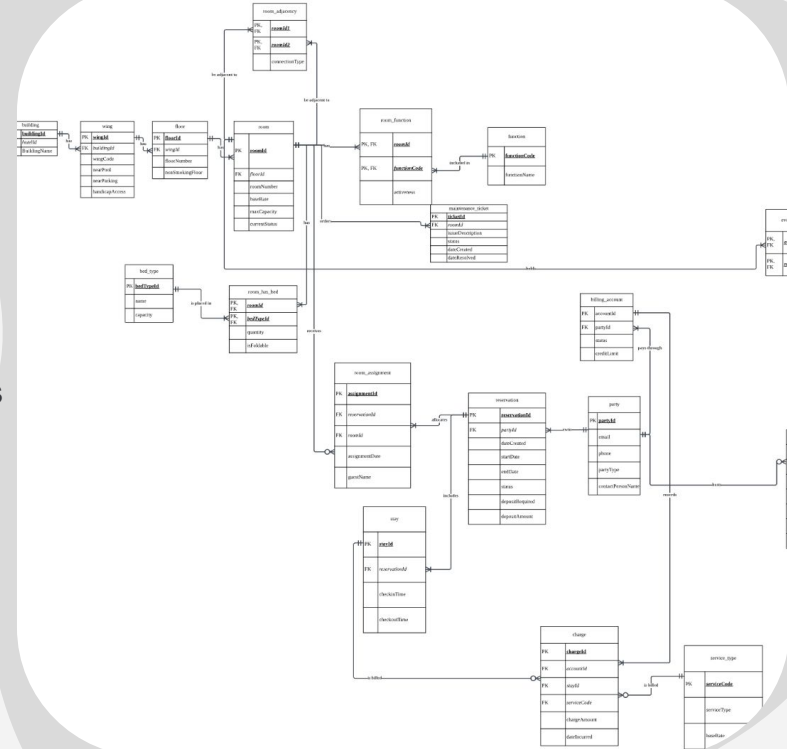
# LAST RESORT HOTEL ADMIN

By Group 9  
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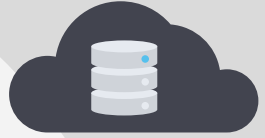


## 5 Logical Sections:

- Group-9/documents/Group9 ERD.pdf at main  
• Annandy2008/Group-9



# SCHEMA DESIGN: ERD → Schema Implementation



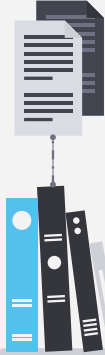
## Core table example:

- **Database:** Use relational database **SQLite**
- **Total Tables:** Created 19 tables in create.sql to ensure data integrity
- **Key Modules:** Management of Hotel Structure, Room Functions, Reservations, and Financial Records

```
CREATE TABLE room (  
    roomId TEXT NOT NULL PRIMARY KEY,  
    floorId INTEGER NOT NULL,  
    roomNumber TEXT NOT NULL,  
    baseRate REAL NOT NULL,  
    maxCapacity INTEGER NOT NULL,  
    currentStatus TEXT NOT NULL,  
    FOREIGN KEY (floorId) REFERENCES floor(floorId),  
    UNIQUE (floorId, roomNumber)  
);
```

## From ERD to SQL:

- **Database Engine:** SQLite
- **Data Integrity:** Enforced via Foreign Keys and Constraints (CHECK, UNIQUE).
- **Normalization:** 3NF design to minimize data redundancy.



# SCHEMA DESIGN – STRUCTURAL POPULATION



## Establishing the Foundation:

**Facility Hierarchy:** Created 2 main buildings: 'Main House' and 'Lakeside Annex' with 6 specific floors and wings.

**Service Catalog:** 6 defined service categories including Sleeping Rooms, Meeting Room Rentals, and Spa, etc.

**Functional Mapping:** Rooms are assigned specific roles (e.g., GL-A-101 from Garden Lodge Building as 'SLP', which is sleeping room).

```
INSERT OR IGNORE INTO building (buildingId, hotelId, buildingName) VALUES  
(3, 2, 'Main House'),  
(4, 2, 'Lakeside Annex');
```

```
INSERT OR IGNORE INTO service_type (serviceCode, serviceType, baseRate) VALUES  
(7, 'Parking', 18.00),  
(8, 'Room Service', 35.00);
```

# SCHEMA DESIGN – COMPLEX BUSINESS DATA



## Simulating Real-World Operations:

- **Multi-Entity Parties:** Managing both individual guests (e.g., Alice Chen) and corporate clients (Acme, Globex, EventCo).
- **Event Management:** Successfully linked Conferences and Weddings to specific room capacities.
- **Inventory Logic:** Tracking complex room features like 'Connecting Doors' and 'Movable Walls'.

## Data Volume at a Glance:

- 20 Rooms with dynamic status (Available, Occupied, Maintenance).
- 180+ Completed/Confirmed Stay records.
- 30 Individual Charge transactions linked to Billing Accounts.





# QUERIES & DATA ANALYTICS

## Main Purposes:

### Support Daily Operations:

Enable staff to perform real-time room searches and guest transactions.

### Data-Driven Management:

Provide a live snapshot of hotel health and financial status.

## How We Structure Queries

### (Some examples):

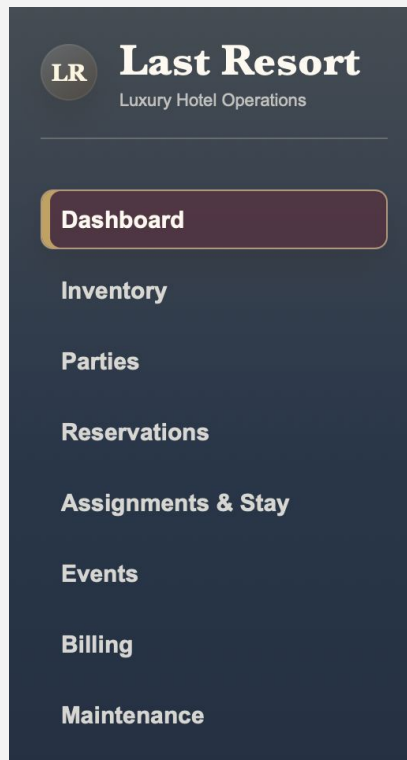
#### Multi-Dimensional room

**Availability:** Filtering rooms by time, location, and guest type.

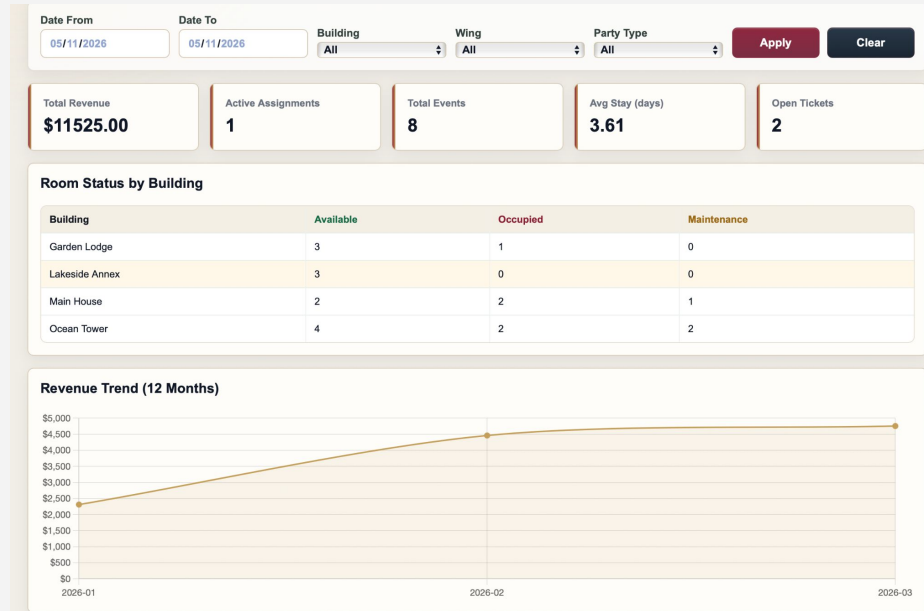
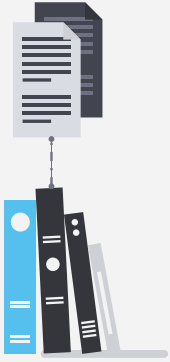
**Inventory Search:** Interactive filtering by physical room attributes and pricing.

#### Real time Billing Aggregation:

Real-time settlement and charge tracking.



# Multi-Dimensional room Availability



**Technical Implementation:**

**Joins:**  
room-floor-wing-building.

**Real-time KPI Aggregation:**  
Instant calculation of Total Revenue, Active Assignments, and Maintenance Alerts (SUM and COUNT logic).

## Query Features:

**Temporal & Spatial Filters:** Search by Date Range, Building, and Wing.

**Contextual Filtering:** Filter by Party Type (Individual vs. Organization) to tailor results.

**Chart:** A revenue chart of recent 12 months that is adjusted according to the filters above

Search & Filter Rooms

Room ID

Status

Function

Bed Type

Max Base Rate (\$)

OT-B

-- All Status --

-- All Functions --

-- All Bed Types --

Up to...

Apply

Clear

Rooms (4 found)

| Room     | Building    | Wing | Floor | Rate     | Capacity | Status      | Non-Smoking |
|----------|-------------|------|-------|----------|----------|-------------|-------------|
| OT-B-101 | Ocean Tower | B    | 1     | \$300.00 | 6        | Available   | No          |
| OT-B-102 | Ocean Tower | B    | 1     | \$320.00 | 8        | Maintenance | No          |
| OT-B-201 | Ocean Tower | B    | 2     | \$280.00 | 4        | Occupied    | Yes         |
| OT-B-202 | Ocean Tower | B    | 2     | \$295.00 | 4        | Maintenance | Yes         |

Update Room Basics

Select Room

OT-B-101

Base Rate

Max Capacity

Status

Add Room Function

Select Room

OT-B-101

Function

Fitness Studio

Activeness

Active

Add Bed Configuration

Select Room

OT-B-101

Bed Type

Bunk Bed

Quantity

1

☐ Foldable Bed

Technical Implementation:

- **Dynamic SQL Generation:** The query adjusts based on which filters are selected (Status, Function, Bed Type).
- **Attribute Mapping:** Direct lookup from `room_function` and `bed_type` catalog tables.

# Inventory Search



Filtering Criteria:

- **Room Status:** Available, Occupied, or Maintenance.
- **Physical Attributes:** Filter by Function (Suite/Meeting) and Bed Type (King/Twin/Sofa).
- **Price Sensitivity:** Max Base Rate filtering for budget-conscious guests.

Update Room Basics

Select Room

OT-B-101

Base Rate

Max Capacity

Status

Available

Add Room Function

Select Room

OT-B-101

Function

Fitness Studio

Activeness

Active

Add Bed Configuration

Select Room

OT-B-101

Bed Type

Bunk Bed

Quantity

1

☐ Foldable Bed

Update Room

Add Function

Add Bed Setup

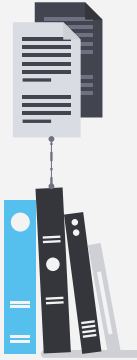
Current Room Functions

| Room     | Code | Name           | Status |
|----------|------|----------------|--------|
| GL-A-101 | SLP  | Sleeping Room  | Active |
| GL-A-102 | SLP  | Sleeping Room  | Active |
| GL-B-101 | GYM  | Fitness Studio | Active |
| GL-B-101 | MTG  | Meeting Room   | Active |
| GL-B-102 | MTG  | Meeting Room   | Active |

Current Bed Configurations

| Room     | Type     | Qty | Foldable |
|----------|----------|-----|----------|
| GL-A-101 | Queen    | 1   | No       |
| GL-A-102 | King     | 1   | No       |
| GL-A-102 | Sofa Bed | 1   | Yes      |
| GL-B-101 | Bunk Bed | 5   | No       |
| GL-B-102 | Bunk Bed | 6   | No       |





# Process of a New Party: New Account, Reservation and Assignments

## Party Management

### Search

Keyword

Party Type

All

Search

### Create Party

Contact Person

Email

Phone

Party Type

Guest

Create

### Add Group Member

Party

31 - Matt Murdock

Guest Name

Add Guest

### Party List

| ID | Type  | Contact      | Email          | Phone    | Account | Account Status | Credit Limit |
|----|-------|--------------|----------------|----------|---------|----------------|--------------|
| 31 | Guest | Matt Murdock | mm6419@nyu.edu | 121-1091 | 33      | Open           | -            |



# Process of a New Party: New Account, Reservation and Assignments

## Reservation Management

View & update

Search and change status

New booking

Party, guests, dates

### Search Reservations

Search by Contact Name...

Search

### Reservation List

| ID | Contact                | Dates                    | Current Status | Update Status                                    |
|----|------------------------|--------------------------|----------------|--------------------------------------------------|
| 68 | Matt Murdock<br>Guest  | 2026-05-12 to 2026-05-29 | Confirmed      | Confirmed <span>↓</span> <button>Update</button> |
| 67 | Mei-Ling Zhou<br>Guest | 2026-05-01 to 2026-05-08 | Completed      | Completed <span>↓</span> <button>Update</button> |

## Assignments & Stay Management

### Actionable Reservations (Arrivals & Current Guests)

| Reservation ID | Guest / Party | Dates                    | Room Assigned | Status / Action |
|----------------|---------------|--------------------------|---------------|-----------------|
| 68             | Matt Murdock  | 2026-05-12 to 2026-05-29 | MH-B-101      | Checked In      |

### Assignment History

| Res ID | Party         | Room     | Date       |
|--------|---------------|----------|------------|
| 68     | Matt Murdock  | MH-B-101 | 2026-05-12 |
| 67     | Mei-Ling Zhou | MH-B-102 | 2026-04-29 |
| 66     | andy anderson | GL-B-101 | 2026-04-28 |
| 65     | Mei-Ling Zhou | OT-A-102 | 2025-05-29 |
| 64     | James Adeyemi | OT-A-101 | 2025-05-09 |

### Current & Past Stays

| Stay ID | Party         | Check-In            | Check-Out / Action             |
|---------|---------------|---------------------|--------------------------------|
| 184     | Matt Murdock  | 2026-05-11 23:06:52 | <button>Check-Out Now</button> |
| 183     | Mei-Ling Zhou | 2026-04-29 13:42:39 | 2026-04-29 13:42:53            |
| 182     | andy anderson | 2026-04-27 22:25:33 | 2026-04-27 22:26:57            |
| 10      | andy anderson | 2026-04-26 13:42:50 | 2026-04-26 13:42:54            |
| 9       | Charlie Puth  | 2026-04-26 13:42:40 | 2026-04-26 13:42:55            |



# Real time Billing

## Billing & Settlement

### Select Account to Manage

| ID  | Guest            | Action                     |
|-----|------------------|----------------------------|
| #33 | Matt Murdock     | <button>View Bill</button> |
| #32 | Mei-Ling Zhou    | <button>View Bill</button> |
| #31 | andy anderson    | <button>View Bill</button> |
| #30 | Ingrid Larsson   | <button>View Bill</button> |
| #29 | Kenji Nakamura   | <button>View Bill</button> |
| #28 | Fatima Al-Rashid | <button>View Bill</button> |
| #27 | Raj Sharma       | <button>View Bill</button> |

### Create New Account

Party

Status

Create

### Invoice for: Mei-Ling Zhou (Account #32)

#### 1. Room Charges (Auto-calculated)

|                               |          |
|-------------------------------|----------|
| Stay #25 (OT-A-102) - 3 days  | \$660.00 |
| Stay #45 (OT-A-102) - 3 days  | \$660.00 |
| Stay #65 (OT-A-102) - 3 days  | \$660.00 |
| Stay #183 (MH-B-102) - 1 days | \$170.00 |

#### 2. Event Charges (Auto-calculated)

No events hosted.

#### 3. Other / Manual Charges

| Date       | Service | Stay Ref | Amount   |
|------------|---------|----------|----------|
| 2026-04-29 | Parking | -        | \$180.00 |

Total Due: **\$2330.00**

### Add 'Other' Charge for Mei-Ling Zhou

Related Stay (Optional)

Service Type

Amount

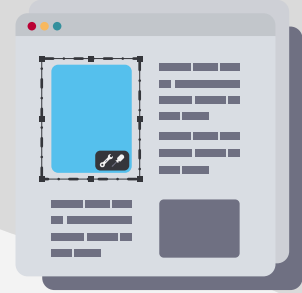
Date Incurred

Add This Charge

## Technical Implementation:

**Update Logic:** Transactional queries ensure data consistency across **stay** and **room** tables.

**Account Joins:** **billing\_account** ⋈ **charge** ⋈ **service\_type**.



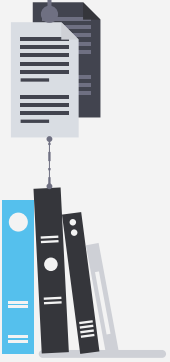
## Real-Time Billing:

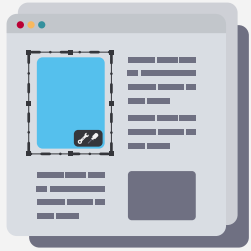
**Account Aggregation:** Summarizes all **charges** associated with a **billing\_account**.

**Service Transparency:** Breaks down costs by Service Type (Room Rental, Spa, Restaurant).

# Demo

<https://edstem.org/us/courses/91931/workspaces/pvgmzOZm9c4hQWApOhAjon6h8wxjaSPZ>





# Thanks!

Do you have any questions?

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